

Overview of work for the past week

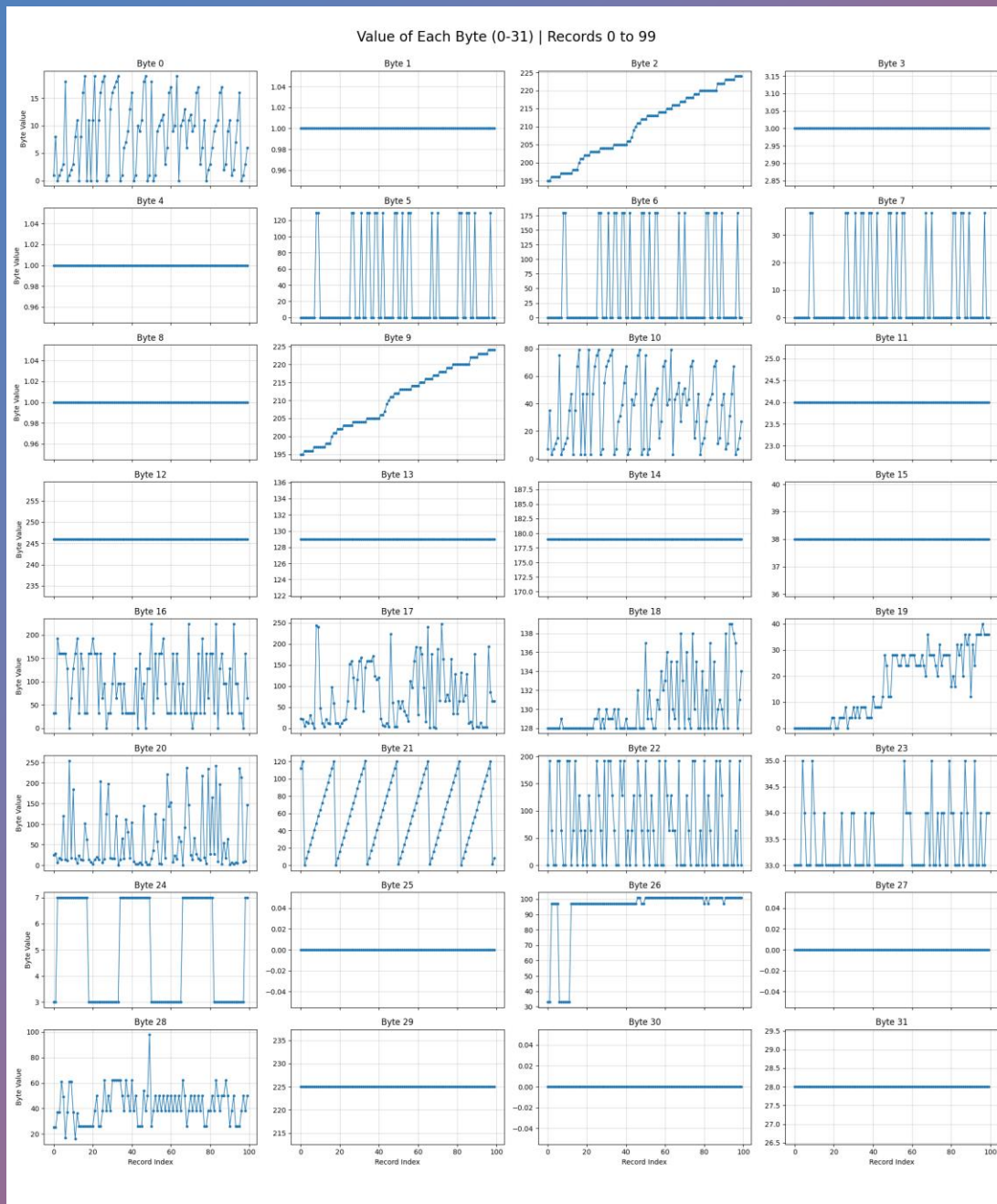
- Logging (more) logs
- The curious case of lajja didi's logs
- Blast from the past
- Coping with B888
- Admitting Defeat :(
- In the future: Asking for help :)

Logging to detect Carrier Aggregation

- I collected more logs to find ascertain the CA bit, but none of my logs had any carrier aggregation, which is strange. The few samples that did contain some traces of CA turned out to be 4G plus (so the point that I had submitted last week about failed sanity check was bogus).
- I spoke to Sapna ma'am about this and she told me that some samples from Lajja Didi's house had CA set to true, and that I should probably look at them.
- I checked and found that most samples collected on the first day had carrier aggregation active while none of the samples that were collected on the second day had any ca, which tells me that I need be more persistent with sample collection. I'd just collect a few from one location and move on to the next. Although I am still not sure why this is so, like why does it vary so much from day-to-day.

Auditing the logs collected at lajja didi's house

- B887 audit to locate the CA bit:
 - The Good news: The fifth byte, where I was expecting the CA bit, never deviates from values 0 and 128 in over 400,000 records.
 - The Bad news: The first two bits of byte five where I was expecting the carrier ID to be located, based on the info from b883 logs, are always zero.
- Recap from previous semester: bytes 5, 6, and 7 either all zero or 81 b3 26 similar to NR-ARFCN (I had forgotten this as well).
- Once again concluded that the above holds true. I don't know what this implies but the 2 bit right-shifted value of the three bytes is 634080 which is the frequency for Band n78. My hypothesis (which are frequently wrong) is that if indeed CA is happening then this must be the secondary cell frequency.
 - Expectation: $(\text{NR-ARFCN} \ll 2) \mid (\text{Carrier_ID} \& 0x03) \rightarrow \text{Little Endian [Byte 5, Byte 6, Byte 7]}$
 - Reality: $(\text{NR-ARFCN} \ll 2) \rightarrow \text{Little Endian [Byte 5, Byte 6, Byte 7]}$



Recall

- Bytes 5, 6, and 7 all vary together, that is when one is non-zero the others are non-zero as well. Otherwise, they are all non-zero with uniform but different magnitudes.

Evaluating a new log type b888

- When I was not making progress, I wondered what sir would suggest, probably to look at the MobileInsight repo, so that is what I did.
- There I found a new log type b888 (NR_MAC_PDSCH_Stats), and the very first value of a b888 record is carrier ID, so I enthusiastically wrote a parser for b888 on the lines of b887 (gemini did most of the work, I just proof-read it).
- When I tried to port the parser to version 3 (using lajja didi's logs as the test driver), I ran into my next problem the first thirty-two bits of a record where I would expect the carrier ID to lie do not change at all.

What to do next

- The only indicator for carrier aggregation we have is gnettrack pro, but the output provided by it is not very detailed. Since nothing else is matching I have also grown suspicious of the value reported by gnettrack pro. A good starting point would be to do what sir suggested and find out how gnettrack pro gets to know about Carrier Aggregation.
- As an aside, since I was getting nowhere I took a look at the logs I had collected in the previous semester and found that the three bits are a little more dynamic than what we see in the logs collected at lajja didi's house, but we were not using gnettrack pro back then so I cannot say anything conclusively.